

VEDANTH ANIL WARRIER

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PROFESSIONAL SUMMARY

Second-year Aerospace Engineering student at UNSW (WAM 88.9, ATAR 99.65) with hands-on experience in SolidWorks FEA, fabrication, embedded systems, and Python numerical simulation of orbital dynamics and spacecraft attitude control. Builds simulation tools from first-principles physics and validates hardware designs through FEA and physical testing.

EDUCATION

UNSW Sydney

Bachelor of Engineering (Hons)

Kensington, Sydney

2025 – Present

- **Major:** Aerospace Engineering
- **Current WAM:** 88.929 (High Distinction Average)
- **Academic Record:** Engineering Mathematics 2E (96), Mechanics of Solids (93), Computing 1A - C Programming (92), Numerical Methods & Statistics (93), Design (91), Engineering Mechanics (90), Fluid Mechanics (89), Electrical Circuits (85).

Normanhurst Boys High School

Higher School Certificate (HSC) | ATAR: 99.65

Normanhurst, Sydney

2019 – 2024

ENGINEERING PROJECTS

ADCS Simulator

Backend & Physics Developer

Personal Project

Aug 2026 – Oct 2026

- **Rigid-Body Dynamics:** Modelled three-axis satellite attitude as a 7-state system (quaternion kinematics and Euler's rigid body equations with reaction wheel gyroscopic coupling), which was integrated with `scipy.integrate.solve_ivp` to reproduce torque-free precession and correct attitude of an asymmetric body.
- **Attitude Control:** Implemented a quaternion-error PD controller with inertia-scaled gains and a feed-forward term cancelling body and wheel gyroscopic coupling, run as a 20 Hz discrete loop under per-axis wheel RPM and spin-up rate limits.
- **Saturation & Deployment:** Modelled per-axis time-to-saturation under constant disturbance torque, and served both simulation modes through a Flask REST API deployed on Render.

Orbital Trajectory & Telemetry Suite

Solo Developer

Personal Project

May 2026 – Jun 2026

- **IVP Trajectory Simulator:** Numerically integrated a **5-state ODE system** (position, velocity, mass) in polar coordinates using `scipy.integrate.solve_ivp` (RK45), modelling gravity-turn ascent with **dynamic atmospheric drag** interpolated from a real altitude-density dataset across **1000 km** of scale height.
- **BVP Orbital Insertion Solver:** Formulated a **3-DOF boundary value problem** to solve for circular orbit insertion parameters across a **3-phase mission** (ascent burn, coast, apogee kick), solved using `scipy.optimize.least_squares` with dynamic variable scaling.

Go-Kart Engineering & Fabrication

Designer & Fabricator

Personal Project

May 2025 – May 2026

- **Design & Simulation:** Engineered the primary chassis and battery holding beams using SolidWorks static FEA, validating an inflated 150kg occupant load and a 100kg battery allowance (Load Factor > 2). Confirmed a **Factor of Safety > 2** (peak stress of 120 MPa against a 250 MPa yield strength) and restricted maximum deflection to 0.9cm, preserving a safe 20cm operational ground clearance.
- **Manufacturing & Assembly:** Fabricated custom chassis and sheet metal enclosures using **Gasless MIG Welding**, and soft soldering for the integration and wiring of six 12V 22Ah COTS batteries.
- **Testing & Optimisation:** Conducted physical testing and powertrain optimisation, diagnosing a drivetrain chain-slip fault via slow-motion video analysis achieving an acceleration of **4.5 m/s²**.

OTHER EXPERIENCE

Project Academy

HSC Mathematics Tutor

Sydney

Dec 2024 – Present

- Lead groups, maintain classroom control, and break down complex mathematical concepts for senior students.

TECHNICAL SKILLS

Engineering Software & Programming

- **CAD & FEA:** SolidWorks (Proficient in static FEA, complex assemblies, and 3D component modelling)
- **Numerical Analysis:** Python (NumPy, SciPy, Flask; Linear systems, root-finding algorithms, IVPs, BVPs, and data visualisation)
- **Embedded Systems:** C/C++ (Arduino control logic, mechatronic actuation, and hardware-software integration)

Hardware & Fabrication

- Gasless MIG Welding, Soft Soldering, Surface Mount (SMD) Soldering.

COMMUNITY & FITNESS

- Ultimate Frisbee & Badminton Team Captain (2022–2024); Push-Up Challenge and World's Greatest Shave participant.